

Cross-Layer Macro Regimes

Which macro conditions are associated with stronger or weaker future equity market performance?

This study evaluates whether a four-layer macro regime framework — Liquidity, Credit, Positioning, and Volatility — can classify market environments that lead to different forward equity outcomes.

Key Finding

The model found that future equity performance differs significantly across macro regimes.

The strongest reliable result came from **Regime C**, defined as an unstable macro environment where **two macro layers are showing stress or deterioration**.

Regime C produced the strongest reliable forward equity performance:

Metric	Regime C Result
20D Forward ES Return	4.20%
60D Forward ES Return	10.10%
60D Forward Sharpe Ratio	2.03
60D Forward Max Drawdown	-3.60%

Robustness testing confirmed that Regime C's outperformance survived:

- Independent forward return recalculation
- Bootstrap confidence intervals
- Non-overlapping return samples
- Newey-West / HAC regression adjustment
- Lagged regime testing
- Outlier sensitivity testing
- Subperiod stability testing
- Placebo simulation

Main Conclusion

Cross-layer macro conditions appear to contain statistically significant and economically meaningful information about future equity performance.

The relationship is not simply:

Stable regime = best performance
Crisis regime = worst performance

Instead, the strongest reliable forward equity performance occurred during **unstable but not fully crisis-level conditions**.

Regime C appears to represent a market environment where macro stress is visible enough to create repricing opportunities, but not severe enough to produce full systemic breakdown.

THE PROBLEM THIS SOLVES

Most market analysis relies on one macro indicator at a time.

Common examples include:

- Liquidity
- Credit spreads
- Volatility
- Positioning
- Yield curve
- Market breadth
- Central bank policy
- Economic growth

The problem is that no single macro indicator fully explains market behavior.

Markets often send mixed signals.

For example:

- Liquidity may be supportive while volatility is elevated.
- Credit may remain contained while positioning becomes crowded.
- Volatility may normalize while liquidity deteriorates.
- One macro layer may break before the others confirm.

This creates a practical problem for traders, investors, and market researchers:

How do we classify the market when different macro signals disagree?

This study attempts to solve that problem by combining four macro layers into one regime framework.

Four Macro Layers

Layer	Purpose
Liquidity	Measures whether capital conditions are supportive, flat, or deteriorating
Credit	Measures financial stress and credit-market risk
Positioning	Measures whether investors are neutral, crowded, bullish, or bearish
Volatility	Measures whether risk conditions are calm, elevated, or crisis-like

Each layer is classified as either Normal, Warning, or Stress.

The number of destabilized layers produces a total destabilization score.

That score is mapped into four regimes.

Regime Framework

Regime	Description	Destabilization Score
Regime A	Stable	0
Regime B	Warning	1
Regime C	Unstable	2
Regime D	Crisis	3–4

The objective is not merely to identify the current macro regime.

The objective is to test whether these regimes historically mattered for future equity returns, volatility, drawdowns, and trend quality.

DATA & METHODOLOGY

Framework

The model classifies each historical observation into one of four macro regimes using the combined state of Liquidity, Credit, Positioning, and Volatility.

The destabilization score is calculated by counting how many layers are in Warning or Stress condition.

Regime Definitions

Regime A — Stable

All macro layers are normal.

This represents the lowest destabilization state.

Regime B — Warning

One macro layer is showing deterioration.

This represents early-stage macro instability.

Regime C — Unstable

Two macro layers are showing deterioration.

This represents a broader but not yet crisis-level destabilization environment.

Regime D — Crisis

Three or four macro layers are destabilized.

This represents severe cross-layer stress.

Performance Metrics Tested

For each regime classification date, future equity performance was measured using E-mini S&P 500 futures.

Metrics tested:

- 5D forward ES return
 - 20D forward ES return
 - 60D forward ES return
 - 20D realized volatility
 - 60D forward maximum drawdown
 - 60D forward Sharpe ratio
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Statistical Testing

The study used ANOVA to test whether average outcomes differed across macro regimes.

All tested metrics showed statistically significant differences across regimes.

Metric	ANOVA p-value
ES 5D Forward Return	7.30e-10
ES 20D Forward Return	1.18e-38
ES 60D Forward Return	1.96e-83
ES 20D Realized Volatility	2.51e-238
ES 60D Forward Max Drawdown	4.81e-10
ES 60D Forward Sharpe	1.33e-20

These results indicate that the macro regime classifications are not random groupings.

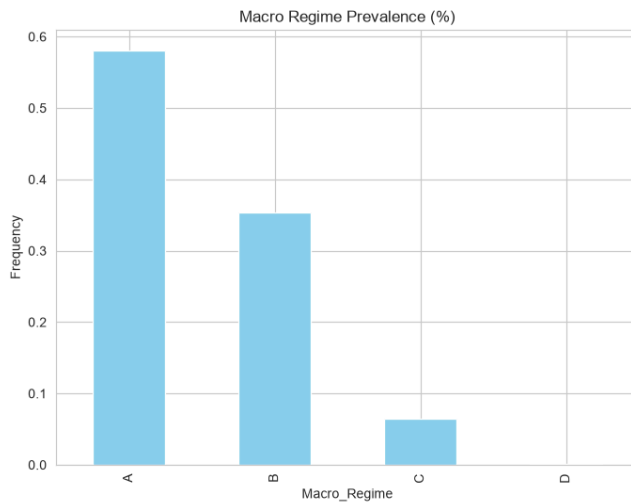
Future equity behavior differed meaningfully across the four regime states.

PAGE 4 — REGIME SUMMARY & KEY CHARTS

Regime Summary

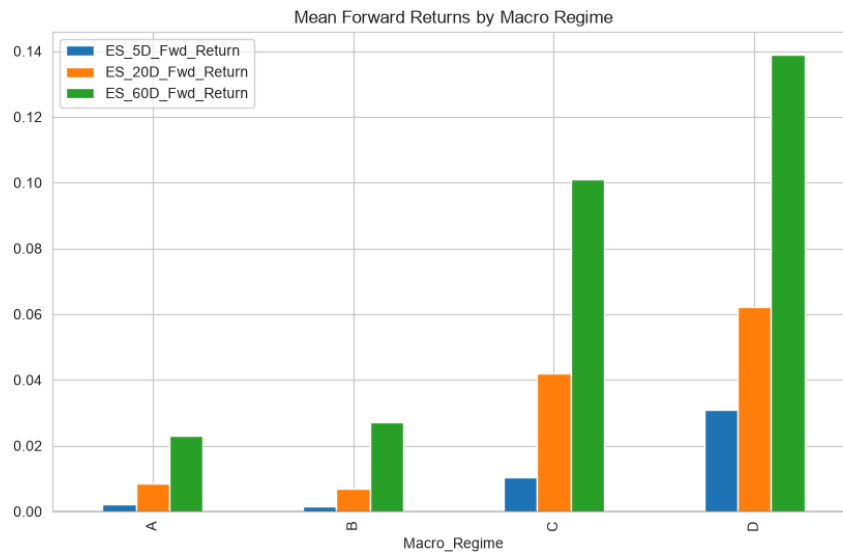
Regime	Observations	Prevalence	20D Fwd Return	60D Max DD	20D Realized Vol
Regime A	2,401	58.05%	0.84%	-4.10%	12.46%
Regime B	1,462	35.35%	0.70%	-5.18%	15.79%
Regime C	267	6.46%	4.20%	-3.60%	30.75%
Regime D	6	0.15%	6.23%	-2.70%	24.69%

Chart - Macro Regime Prevalence



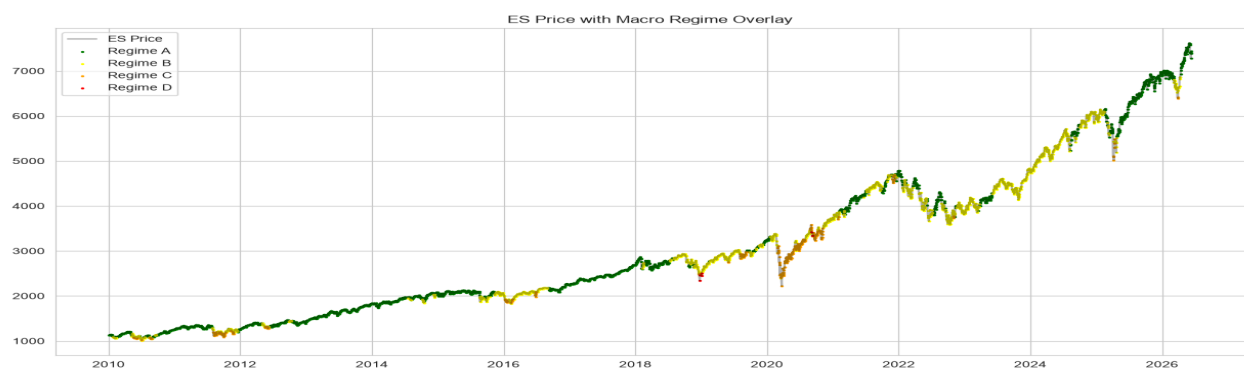
Regime A dominated the sample at 58.05%, followed by Regime B at 35.35%. Regime C occurred less frequently at 6.46%, but still had enough observations for analysis. Regime D occurred only 6 times and should not be used for strong statistical inference.

Chart - Forward Returns By Macro Regime



Regime C stood out as the strongest reliable regime, especially over the 20D and 60D horizons. Regime D is statistically unreliable with only 6 samples.

Chart - Macro Regime Timeline



This chart helps determine whether regime classifications align with meaningful market environments rather than appearing randomly distributed.

PERFORMANCE RESULTS

Forward Performance by Regime

Metric	Regime A	Regime B	Regime C	Regime D
5D Fwd Return	0.21%	0.16%	1.04%	3.11%
20D Fwd Return	0.84%	0.70%	4.20%	6.23%
60D Fwd Return	2.30%	2.71%	10.10%	13.89%
60D Fwd Sharpe	1.24	0.96	2.03	3.31

Important Interpretation

Regime D had the highest raw forward return and Sharpe values.

However, Regime D only had **6 observations**.

That sample size is too small to treat Regime D as a reliable result.

Therefore, the strongest defensible finding is:

Regime C was the highest-performing reliable macro regime.

Regime C produced:

- Approximately 5.0x the 20D return of Regime A
- Approximately 6.0x the 20D return of Regime B
- Approximately 4.4x the 60D return of Regime A
- Approximately 3.7x the 60D return of Regime B

This suggests that the market may perform best when macro stress is broad enough to trigger repricing, but not severe enough to represent full crisis conditions.

Drawdown Profile

Regime	60D Forward Max Drawdown
Regime A	-4.10%
Regime B	-5.18%
Regime C	-3.60%
Regime D	-2.70%

Regime B produced the deepest forward drawdowns.

Regime C produced better average drawdown characteristics than Regime B and broadly competitive drawdown characteristics versus Regime A.

However, bootstrap confidence intervals suggest that Regime C’s drawdown advantage over Regime A should not be overstated.

Chart - Macro Regime Transition Probabilities

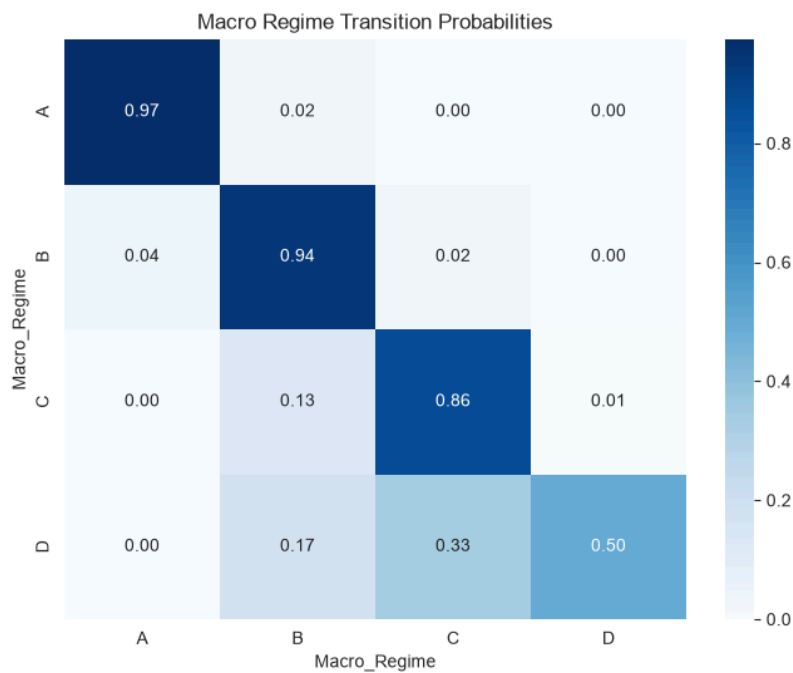
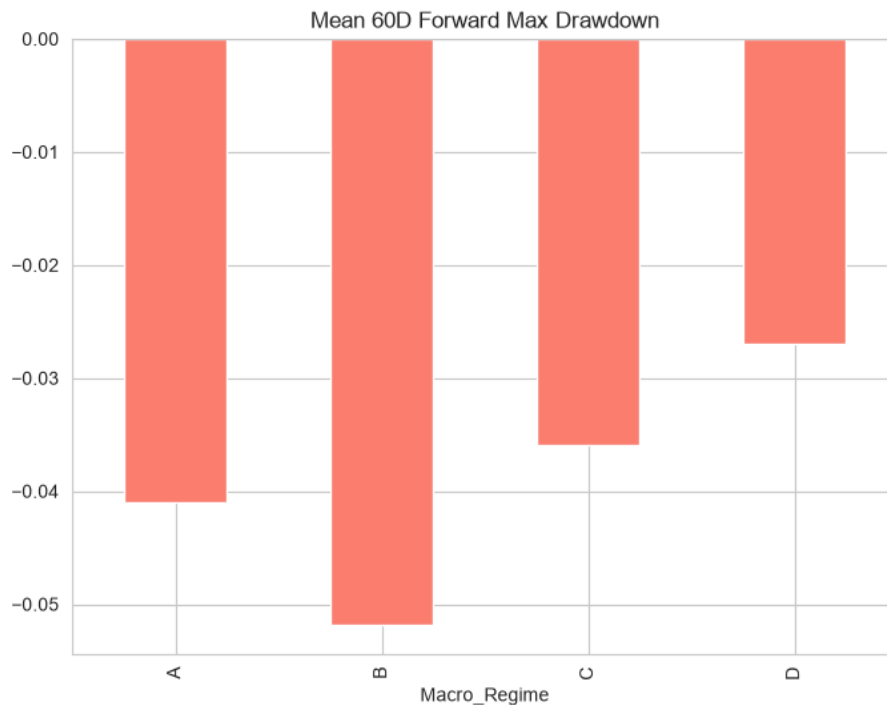


Chart - Drawdowns By Macro Regime



Regime B appears to be the weakest transition environment, with the deepest average forward drawdowns.

ROBUSTNESS & VALIDATION

A separate validation script was used to verify whether the Regime C conclusion survived calculation checks, statistical adjustments, and robustness testing.

The final validation status was that the conclusion is robust.

The dataset passed core integrity checks.

- Date index sorted: True
- Duplicate dates: False
- Largest date gap: 5 calendar days
- Regime mapping mismatches: 0

Forward returns were independently recalculated.

The maximum difference between the original and recomputed values was approximately:

- 5D forward return max difference: 9.97e-17
- 20D forward return max difference: 9.97e-17
- 60D forward return max difference: 9.97e-17

This confirms that the original forward return calculations were correct.

Bootstrap Confidence Intervals

20D Forward Returns

Regime	Mean	5% CI	95% CI
Regime A	0.87%	0.52%	1.24%
Regime B	0.63%	-0.03%	1.23%
Regime C	4.13%	2.73%	5.46%

Regime C's 20D confidence interval remained clearly above Regime A and Regime B.

60D Forward Returns

Regime	Mean	5% CI	95% CI
Regime A	2.30%	1.45%	3.08%
Regime B	2.57%	1.61%	3.46%
Regime C	10.24%	8.51%	12.06%

Regime C maintained a substantially stronger 60D forward return profile.

Drawdown Bootstrap Results

Regime	Mean 60D Max Drawdown	5% CI	95% CI
Regime A	-4.08%	-4.81%	-3.46%
Regime B	-5.20%	-6.21%	-4.29%
Regime C	-3.62%	-5.32%	-2.33%

Regime C had better average forward drawdowns than Regime B.

However, because its confidence interval overlaps with Regime A, the drawdown conclusion should be phrased carefully.

Non-Overlapping Sample Test

Because 20D and 60D forward returns overlap when measured daily, non-overlapping samples were tested.

20D Non-Overlapping Forward Returns

Regime	Mean	Sample Size
Regime A	0.85%	121
Regime B	0.63%	69
Regime C	3.62%	17

Regime C remained the strongest regime in the 20D non-overlapping sample.

60D Non-Overlapping Forward Returns

Regime	Mean	Sample Size
Regime A	1.67%	36
Regime B	3.09%	27
Regime C	8.53%	6

Regime C also remained the strongest in the 60D non-overlapping sample, although the sample size was small and should be interpreted cautiously.

Newey-West / HAC Regression

To adjust for serial correlation caused by overlapping forward returns, a Newey-West / HAC regression was run using Regime A as the base category.

Variable	Coefficient	P-Value	Interpretation
Regime B	+0.41%	0.648	Not statistically different from Regime A
Regime C	+7.80%	<0.001	Statistically significant outperformance
Regime D	+11.59%	<0.001	Statistically significant but unreliable due to only 6 observations

The Regime C coefficient remained highly significant after overlap-adjusted standard errors.

Lagged Regime Test

A 3-day lag was applied to the regime classification to test whether the result survived more realistic reporting conditions.

Metric	Result
Original Regime C 20D Return	4.20%
3-Day Lagged Regime C 20D Return	4.08%

The signal remained largely intact after applying the lag.

Outlier Sensitivity

Regime C remained strong after winsorization.

Version	Regime C 60D Mean
Raw	10.10%
Winsorized	9.50%

This suggests that Regime C's outperformance was not driven entirely by extreme outliers.

Subperiod Stability

Regime C remained the strongest 20D forward return regime across all tested subperiods.

Period	Regime A	Regime B	Regime C
2010–2014	0.91%	0.52%	4.37%
2015–2019	0.87%	0.49%	3.00%
2020–2026	0.70%	0.86%	4.46%

This indicates that Regime C was not merely a single-period anomaly.

Placebo Test

A block-shuffle placebo test was used to compare the actual Regime C result against randomized regime assignments.

Placebo p-value: 0.0000

This indicates that Regime C's observed performance was highly unlikely to occur from random regime labeling.

INTERPRETATION & LIMITATIONS

Interpretation

The results suggest that cross-layer macro regime classification can add value as a market environment filter.

The most important finding is that performance does not improve linearly as the macro environment becomes more stable.

Instead, the data suggests a more nuanced structure:

Regime A — Stable

Regime A is the most common environment.

It produced:

- 0.84% average 20D forward return
- 2.30% average 60D forward return
- 12.46% 20D realized volatility
- 1.24 60D forward Sharpe

This is a normal, constructive environment, but not the highest-return environment.

Regime B — Warning

Regime B appears to be the weakest transition state.

It produced:

- 0.70% average 20D forward return
- 2.71% average 60D forward return
- -5.18% average 60D forward max drawdown
- 0.96 60D forward Sharpe

This suggests that early macro deterioration may create a poor risk-adjusted market environment.

One layer is breaking, but the market may not yet have fully repriced the risk.

Regime C — Unstable

Regime C was the strongest reliable regime.

It produced:

- 4.20% average 20D forward return
- 10.10% average 60D forward return
- -3.60% average 60D forward max drawdown
- 2.03 60D forward Sharpe

Although volatility was elevated at 30.75%, the forward return profile was substantially stronger.

This may indicate that once macro stress becomes visible across two layers, markets begin to reprice quickly, creating stronger forward recovery or continuation opportunities.

Regime D — Crisis

Regime D showed the highest raw forward returns, but only occurred 6 times.

Because the sample size is too small, it should be treated as anecdotal rather than conclusive.

The correct interpretation is:

Regime D may represent extreme dislocation, but more observations are required before drawing reliable conclusions.

Limitations

1. Regime D sample size is too small

Regime D has only 6 observations.

Its results should not be used for strong inference.

2. Macro data may be lagged

Some macro indicators, especially liquidity and positioning data, may be reported with a delay.

Forward filling helps align the dataset, but real-time availability may differ from historical backtest availability.

3. Equal layer weighting may be too simple

The destabilization score gives equal weight to Liquidity, Credit, Positioning, and Volatility.

In reality, some layers may matter more than others.

Credit and Liquidity, for example, may have asymmetric market impact compared with Positioning.

4. Regime definitions are rule-based

The regimes are created from a scoring framework rather than learned directly from a model.

This improves interpretability but may miss nonlinear relationships.

5. This is not yet a trading strategy

The study evaluates predictive relationships.

It does not yet test a fully executable strategy with:

- Position sizing
- Volatility targeting
- Transaction costs
- Slippage
- Risk limits
- Live execution assumptions

6. Further validation is still useful

Future work should test:

- NQ confirmation
- SPY / QQQ confirmation
- Additional asset classes
- Rolling-window stability
- Out-of-sample walk-forward testing
- Layer-specific importance
- Real-time publication lag assumptions

Author

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Background

- Former Research Analyst – Institutional Shareholder Services
- Former Senior Research Analyst (M&A) – FactSet
- Equity Investor Since 2015
- Futures/Forex/Crypto Trader Since 2020
- Quantitative Market Researcher

Related Research

- FX Regime Detection
- FX Correlation Regime Predictability Analysis
- Global Macro Regime Analysis
- NQ/ES Liquidity Sweep Studies
- Session Behavior Analytics
- SMT Divergence Research
- Trend Continuation Probability Analysis

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